Project: Worked examples
Title: Rolling bearing rating life
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Equivalent dynamic load of a rolling bearing under radial and axial load, its basic rating life L_{10} in millions of revolutions and in hours at the running speed, and the life modified for reliability, against a required life. P = X·F_r + Y·F_a. L_{10} = (C/P)^p million revolutions, p = 3 for ball and 10/3 for roller bearings. $L_{10}h = L_{10}\cdot 10^6$ revolutions at speed n. $L_{nm} = a_1\cdot L_{10}$, $L_{nmh} = a_1\cdot L_{10}h$. The equation is used for P ≤ 0.5·C.

Ref: ISO 281 §6 — Basic dynamic load rating and basic rating life
Ref: Shigley's Mechanical Engineering Design Ch. 11 — Rolling-contact bearings

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Bearing type (selects the life exponent)	ball		drawing: DRG-GB-211 rev A	any
C	Basic dynamic load rating	25	kN	drawing: DRG-GB-211 rev A	0.1 … 100000 kN
F_r	Radial load	3	kN	spec: LOAD-SCHEDULE-01	0 … 100000 kN
F_a	Axial load	1.5	kN	spec: LOAD-SCHEDULE-01	0 … 100000 kN
X	Radial load factor (bearing catalogue)	0.56		drawing: DRG-GB-211 rev A	0 … 2
Y	Axial load factor (bearing catalogue)	1.5		drawing: DRG-GB-211 rev A	0 … 5
n	Speed	1000	rpm	spec: LOAD-SCHEDULE-01	1 … 100000 rpm
a_1	Life modification factor for reliability	0.64		spec: SPEC-MECH-03 design criteria	0.05 … 1
L_req	Required life (0 for no criterion)	2000	h	spec: SPEC-MECH-03 design criteria	0 … 1000000 h

3. Assumptions

A1. Steady load and speed; the bearing is correctly fitted, lubricated and free of contamination. (Basic rating life conditions.)
A2. X and Y are the catalogue factors for this bearing and load ratio. (Supplied by the user.)

4. Working

`[10] p = 3` ISO 281 §6
`= 3`
Life exponent (ball bearing)

[11] P = X * F_r + Y * F_a ISO 281 §6
= 0.56 × 3 kN + 1.5 × 1.5 kN
= 3.93 kN
Equivalent dynamic load
[12] k_P = P / C
= 3.93 kN / 25 kN
= 0.1572
Load ratio P/C
[13] L_10 = (C / P)^p ISO 281 §6
= (25 kN / 3.93 kN)^3
= 257.42
Basic rating life (millions of revolutions)
[14] L_10h = L_10 * 1e6 * 2 * pi / n
= 257.42 × 1e6 × 2 × π / 1000 rpm
= 4290.3 h
Basic rating life in hours (10^6 revolutions of 2π rad at speed n)
[15] L_nm = a_1 * L_10
= 0.64 × 257.42
= 164.75
Modified rating life (millions of revolutions)
[16] L_nmh = a_1 * L_10h
= 0.64 × 4290.3 h
= 2745.8 h
Modified rating life in hours
[17] u_life = L_req / L_nmh
= 2000 h / 2745.8 h
= 0.72838
Required over modified life

5. Results

SYMBOL	VALUE	DESCRIPTION
P	3.93 kN	Equivalent dynamic load
k_P	0.1572	Load ratio
L_10	257.42	L_{10} (millions of revolutions)
L_10h	4290.3 h	L_{10} in hours
L_nm	164.75	Modified life (millions of revolutions)
L_nmh	2745.8 h	Modified life in hours

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Modified life against required life	u_life = 0.72838	u_life ≤ 1	0.728	ISO 281 §6

7. Applicability

No warnings.

8. Verification

Template mech.bearing-life@1.0.0 Content hash 01f73131b0af3ef6 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0